OmniAI RAG Architecture & System Specification

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1. Overview of Retrieval-Augmented Generation (RAG)
OmniAI combines Google Gemini 3.1 Flash Lite with local Vector Storage.
When a PDF is uploaded, text chunks are converted to vectors using text-embedding-004.
Semantic search retrieves the top 4 chunks with similarity scores.

2. Concurrency and Virtual Threads in Spring Boot 3.5
Virtual threads are lightweight threads managed by the JVM rather than the OS.
OmniAI achieves throughput of over 10,000 requests per second with Virtual Threads.

3. OmniAI Security Policies and Data Protection

All document vectors are kept in RAM memory using SimpleVectorStore.

No sensitive document data is retained permanently after the clear command.

The recommended chunk size is 800 tokens with a 100 token overlap.

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